

Unconfoundedness I: Assumptions and Matching

BUSS975: Causal Inference in Financial Research

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Outline

1. Introduction: what actually resolves bias
2. Confounders, covariates, and colliders
3. Identification assumptions
4. Subclassification
5. Exact matching
6. Nearest-neighbour matching
7. Inference with matching
8. Bias correction

Lecture 3b takes the same assumption further: collapse X to one number, weight instead of match, then compare with regression.

1. Introduction

Where we left off

- Lecture 2 gave a rule for **which** variables to condition on: block every backdoor path; condition on no descendant of D .
- It was silent on two things:
 - ▶ What if the variables that close the backdoors are **unobserved**? (Usually the case.)
 - ▶ Given a valid set, **how** do we actually condition — and does it matter?
- This lecture takes the optimistic branch: **suppose the confounders are observed**.
- Subclassification, matching, propensity scores and regression are all answers to “how.” They differ less than their names suggest.

Treatment assignment mechanisms resolve bias, not models

Recall Lecture 1's decomposition. It is an **identity** — true in every dataset, experimental or not:

$$\underbrace{E[Y|D=1] - E[Y|D=0]}_{SDO} = ATE + \underbrace{E[Y^0|D=1] - E[Y^0|D=0]}_{\text{selection on levels}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{selection on gains}}$$

- Nothing here is an assumption; it holds whatever you do.
- **A regression cannot delete the last two terms.** No functional form, no fixed effects, no clustering choice touches them.
- What deletes them is a **claim about how treatment was assigned.**

What randomization actually did

- Under $(Y^1, Y^0) \perp\!\!\!\perp D$, we were permitted to *deduce*:

$$E[Y^0|D=1] = E[Y^0|D=0], \quad E[Y^1|D=1] = E[Y^1|D=0]$$

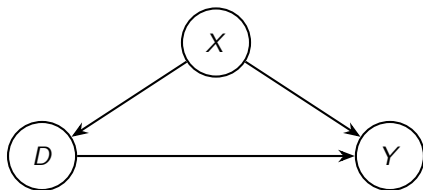
- The first kills selection on levels. Both together kill $ATT - ATU$. What is left is the ATE .
- These came from the **assignment mechanism**, not the estimator. We then *inserted* them into a statistical model.

The statistical model alone cannot resolve bias — the statistical model matched with the assignment mechanism can.

Why this framing matters

- Randomization is an **assignment mechanism**, not a statistical model. A regression is a **statistical model**, not an assignment mechanism.
- Confusing them is the most common error in empirical finance: treating a long control list as an identification strategy.
- Unconfoundedness is this lecture's mechanism: treatment assigned **as if at random within strata of X** .
- Everything that follows is a statistical model, credible *only* if you can defend that mechanism.
- Defend the mechanism first; choose the estimator second.

The bet we are making



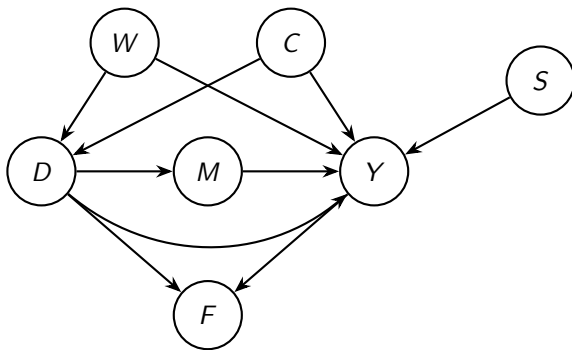
- Everything rests on one claim: **there is no dashed circle**. No unobserved U pointing at both D and Y .
- That claim is **not testable**. No diagnostic, balance table, or specification test can establish it.
- It is defended with institutional knowledge, not statistics.
- The literature calls it **unconfoundedness**, **conditional independence (CIA)**, or **selection on observables**. Same thing.

2. Confounders, Covariates, and Colliders

A worked DAG: first class on the Titanic

- **Question:** did a first-class ticket (D) raise your chance of surviving (Y)?
- Raw comparison: 62.5% of first-class passengers survived versus 27.1% of everyone else — a gap of 35.4 points.
- But the “*women and children first*” rule was actually enforced, and first-class cabins were closer to the boat deck.
- **Sex** and **age** drive both who bought first class and who reached a lifeboat. Textbook confounding.
- The data are real, complete, small enough to compute by hand, and the confounders are undisputed.

The Titanic DAG



D first class Y survival W sex C age M lifeboat F fame S swimming

- Backdoor paths: $D \leftarrow W \rightarrow Y$ and $D \leftarrow C \rightarrow Y$. Both **open**. Condition on $\{W, C\}$ and both close.
- F is a **collider**: $D \rightarrow F \leftarrow Y$. Already closed; conditioning on it *opens* it.
- M is a **mediator**: first class sat nearer the boat deck. It is *part of* the effect, not a confounder.

Four kinds of variable

Type	Example	What conditioning does
Confounder	W sex, C age	Required. Closes a backdoor path.
Collider	F fame	Forbidden. Opens a closed path.
Neutral / precision	S swimming	Optional. No bias either way; shrinks the standard error.
Mediator	M lifeboat	Forbidden for the total effect. Blocks the causal path.

- Only one of these four rows is about bias *reduction*. Two are about bias *creation*.
- S predicts the outcome but not treatment: free precision. The honest case for non-confounder controls.
- M is subtle. Conditioning on it asks: *did first class help, holding fixed the main way it helped?* Near zero — and the wrong question.
- The categories are properties of the **graph**, not the variable. Press coverage *before* boarding would not be a collider.

The same DAG in corporate finance

Effect of a **bank credit line** (D) on **firm investment** (Y):

Titanic	Finance analogue	Rule
W , C sex, age	Firm size, industry, credit rating	Condition on
F fame	<i>Subsequent</i> analyst coverage	Never condition on
S swimming	Pre-period export-market demand	Optional, improves precision
M lifeboat	Amount actually drawn on the line	Never condition on
—	Unobserved managerial quality	The thing that sinks you

- Analyst coverage is the finance collider: it responds to *both* the financing event and firm performance.
- The drawn amount is the mediator a referee will *ask* for — “surely control for how much they borrowed.” That removes the channel you are measuring.
- The last row has no Titanic counterpart, and that is the whole problem. On the Titanic we know the confounders; in finance we rarely do.

Covariate selection without a DAG

In practice you rarely have expert consensus on the full graph. Three working rules:

1. **Include variables you are confident are confounders.** Prior knowledge about the assignment mechanism is the input, not the data.
 2. **Include pre-treatment variables that strongly predict the outcome.** These are confounders or precision-improvers like S , and *probably not colliders* — a variable measured before treatment cannot be caused by it.
 3. **Exclude anything that might be an outcome.** If you are unsure, exclude everything measured after treatment.
- Rule 2 does most of the work; rule 3 is most often violated in finance.
 - “Pre-treatment” does the heavy lifting in rule 2: a timing argument, and timing is checkable.

3. Identification Assumptions

Selecting the target parameter

Before estimating anything, decide *whose* effect you want.

$$ATE = E[Y^1 - Y^0], \quad ATT = E[Y^1 - Y^0 \mid D=1], \quad ATU = E[Y^1 - Y^0 \mid D=0]$$

- A polio vaccine. The three parameters answer three different policy questions.
 - ▶ *ATE*: what if we vaccinate **everyone**?
 - ▶ *ATT*: how much did the vaccine help **those who took it**?
 - ▶ *ATU*: what would happen if we reached **the currently unvaccinated**?
- They coincide only under constant effects. With heterogeneity they differ, sometimes a lot.
- **The estimand is a choice you make about the policy question.**

Which parameter does the question need?

Finance question	Parameter
Should the regulator extend this mandate to all firms?	<i>ATE</i>
Did the firms that adopted IFRS benefit from it?	<i>ATT</i>
Would forcing non-adopters to adopt help them?	<i>ATU</i>

- The third row is the *expansion* question — almost never the one your data answers well.
- Papers routinely estimate the *ATT* then draw conclusions requiring the *ATE* or *ATU*. Watch for it in referee reports — including your own.

Unconfoundedness is a version of independence

- Strong unconfoundedness:

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid X$$

- Compare Lecture 1's randomization assumption, $(Y^1, Y^0) \perp\!\!\!\perp D$, with no conditioning.
- Among units with the *same* X , treatment is as good as randomly assigned: **conditional randomization**.
- The §1 assignment mechanism, stratum by stratum:

$$E[Y^1 \mid X=x] = E[Y^1 \mid D=1, X=x] = E[Y^1 \mid D=0, X=x]$$

$$E[Y^0 \mid X=x] = E[Y^0 \mid D=1, X=x] = E[Y^0 \mid D=0, X=x]$$

- Strong because it's conditional independence with respect to **both** potential outcomes.

What the assumption buys: plug-in

- Y^1 and Y^0 are never both observed.
- Unconfoundedness lets us **replace a missing counterfactual with an observed outcome** from a comparable unit:

$$\underbrace{E[Y^0 \mid D=1, X=x]}_{\text{never observed}} = \underbrace{E[Y^0 \mid D=0, X=x]}_{\text{observed}} = \underbrace{E[Y \mid D=0, X=x]}_{\text{in your data}}$$

- The first equality is the **assumption**. The second is the switching equation — pure definition.
- Every method in this lecture is a different way of executing that substitution.
- Switching estimators cannot rescue a design: all lean on the first equality, none can test it. **Identification** credibility is common to all.
- They differ downstream — functional form, extrapolation, which strata get weight. Those are **estimation** differences; Lecture 3b §3 returns to them.

Unconfoundedness is controversial

- Read literally: firms with identical observables **flip a coin** over a major financial decision.
- Every model of optimizing behaviour says otherwise: firms choose treatment *because* they expect to benefit — Lecture 1's selection on gains.
- This is the deepest objection, and it comes from economics, not statistics.
- Not a reason never to use these methods — a reason to state why *your* setting escapes it.

Guido W. Imbens's three defences

1. **Statistical motivation.** It is the natural starting point.
 2. **It is an empirical question.** With enough institutional knowledge the relevant X may be measurable. Whether it is depends on the setting, not econometrics.
 3. **Objectives differ.** The decision-maker optimizes something *other than your outcome*: a firm choosing a credit line optimizes liquidity and covenant slack, not the investment measure you study.
- Defence 3 is the strongest, and worth naming explicitly.
 - It also says when the design *fails*: when the agent optimizes precisely your outcome.

Road map: two routes through this chapter

Everything ahead assumes unconfoundedness. The methods split on **what they do when a treated unit has no comparable control**.

	Route 1: interpolation	Route 2: extrapolation
Rests on	Common support	Functional form
Fills the gap by	It does not — it refuses	Projecting the fitted model into it
Methods	Subclassification, matching, propensity scores	Regression
Covered in	§4–§8, and Lecture 3b §1	Lecture 3b §2

Common support

- Unconfoundedness alone is not enough. You also need units to compare.
- **Strong common support** (needed for the *ATE*):

$$0 < \Pr(D = 1 \mid X = x) < 1 \quad \text{for all } x$$

At every value of X , both treated and untreated units must exist. <!--

Cunningham's jigsaw analogy: **unconfoundedness is the blueprint; common support is whether the pieces are actually in the box.**

- A perfect blueprint is useless if a piece is missing — and a complete box is useless if the blueprint is wrong. —>

The two assumptions differ in kind

Unconfoundedness	Common support
Untestable	Verifiable in your data
Defended with institutional knowledge	Checked with a histogram
Failure $\Rightarrow$ bias	Failure $\Rightarrow$ extrapolation

- You can have one without the other, and the failures look nothing alike.
- Common support is the assumption you have **no excuse** for not checking — and the one most papers skip.
- This distinction is the key to the NSW debate in Lecture 3b.

Weighting treatment effects under unconfoundedness

Put the two together and the ATE becomes computable.

$$\begin{aligned} ATE(x) &= E[Y^1 | X=x] - E[Y^0 | X=x] \\ &= \underbrace{E[Y^1 | D=1, X=x] - E[Y^0 | D=0, X=x]}_{\text{unconfoundedness — used on both terms}} \\ &= \underbrace{E[Y | D=1, X=x] - E[Y | D=0, X=x]}_{\text{switching equation}} \\ ATE &= \sum_x \underbrace{\left(E[Y | D=1, X=x] - E[Y | D=0, X=x] \right) \Pr(X=x)}_{\text{iterated expectations; common support}} \end{aligned}$$

- $ATE(x)$ is the effect **within stratum** x — the CATE. The last line averages those with **population** weights.
- **Unconfoundedness is used twice**, for Y^1 and for Y^0 . That is why the ATE needs the *strong* version.

A three-stratum example

Suppose the effect on mortality differs by age group, and the groups are equally sized:

Stratum	Effect in stratum (pp)	$\Pr(X = x)$
Young	80	1/3
Middle-aged	55	1/3
Old	100	1/3

$$\begin{aligned}\widehat{ATE} &= \frac{N_y}{N} \times SDO_y + \frac{N_m}{N} \times SDO_m + \frac{N_o}{N} \times SDO_o \\ &= \frac{1}{3} \times 80 + \frac{1}{3} \times 55 + \frac{1}{3} \times 100 \\ &= 78.33\end{aligned}$$

- The *ATE* is a number **no stratum experiences**. It describes a population, not a unit.

Estimating the ATT under weaker assumptions

- A weaker assumption suffices if you only want the *ATT*:

$$Y^0 \perp\!\!\!\perp D \mid X$$

- Only the **untreated** potential outcome must be independent of treatment.
- Firms may *anticipate their own gains* Y^1 and select on them, but not on how they would have done anyway.
 - ▶ Often defensible: a firm knows its expected benefit from a credit line better than its counterfactual path without one.
- **Weak common support:** Controls must exist wherever treated units do — but not conversely.

$$\Pr(D = 0 \mid X) > 0 \quad \text{for all } X \text{ such that } \Pr(D = 1 \mid X) > 0$$

Why the asymmetry makes sense

$$ATT(x) = E[Y^1 \mid D=1, X=x] - E[Y^0 \mid D=1, X=x]$$

$$= \underbrace{E[Y \mid D=1, X=x]}_{\text{switching equation}} - E[Y^0 \mid D=1, X=x]$$

$$= E[Y \mid D=1, X=x] - \underbrace{E[Y \mid D=0, X=x]}_{\text{weak unconfoundedness}}$$

$$ATT = \underbrace{\sum_x \left(E[Y \mid D=1, X=x] - E[Y \mid D=0, X=x] \right) \Pr(X=x \mid D=1)}_{\text{common support, wherever treated units live}}$$

- Weighting by $\Pr(X \mid D=1)$ — covariates **among the treated** — makes it the *ATT*: strata with no treated units get zero weight.
- **Rule**: if you can only defend the weak version, your estimand is the *ATT*. Say so.

4. Subclassification

Cochran's method

- The oldest estimator here, and the most transparent. Three steps:
 1. **Stratify** on X .
 2. Compute the treated-minus-control difference **within** each stratum.
 3. **Average** the strata, weighting by whatever the estimand requires.
- No functional form is assumed. Within a stratum, units are identical on X by construction.
- The weights define the estimand:

Estimand	Weight stratum k by	Interpretation
ATE	n_k / N	population share
ATT	$n_{k,treated} / N_{treated}$	treated share
ATU	$n_{k,untreated} / N_{untreated}$	control share

The Titanic: the raw comparison

- Treated: 325 first-class passengers, 62.5% survived.
- Control: 1,876 others, 27.1% survived.

$$SDO = 62.5 - 27.1 = 35.4 \text{ percentage points}$$

- Taken at face value: a first-class ticket nearly *tripled* your survival odds.
- But first class held a very different mix of people. Sex and age are confounders, and observable.
- Stratify on {sex, age}: four strata, all occupied.

The four strata

Stratum	<i>n</i>	first class	survival, 1st	survival, other	difference
Adult male	1,667	175	32.6%	18.8%	13.7 pp
Adult female	425	144	97.2%	62.6%	34.6 pp
Male child	64	5	100.0%	40.7%	59.3 pp
Female child	45	1	100.0%	61.4%	38.6 pp

- Every within-stratum difference is **smaller than the 35.4 pp raw gap**.
Confounding was inflating the estimate.
- Composition: 76% of passengers were adult males — the *smallest* effect — but only 54% of first class.
- The last stratum holds **one** first-class female child. Remember her.

Three estimands, three answers

Same four differences. Only the weights change.

Weight on	Adult male	Adult female	Male child	Female child
ATE (population share)	0.757	0.193	0.029	0.020
ATT (treated share)	0.538	0.443	0.015	0.003
ATU (control share)	0.795	0.150	0.031	0.023

$$\widehat{ATE} = 19.6 \text{ pp}, \quad \widehat{ATT} = 23.8 \text{ pp}, \quad \widehat{ATU} = 18.9 \text{ pp}$$

- The raw 35.4 pp falls to roughly **20 pp** once composition is held fixed. Nearly half the headline gap was confounding.
- $ATT > ATE$ because first class was disproportionately adult *female* — the stratum with a large effect.

The curse of dimensionality

- Now delete one passenger: the single first-class female child.
- The stratum still exists — 44 girls travelled in other classes — but has **no treated unit**. The difference is undefined.

Parameter	Still identified?
<i>ATE</i>	No — needs all four differences
<i>ATU</i>	No — female children get weight 0.023
<i>ATT</i>	Yes — 23.7 pp from the three occupied strata

- One passenger of 2,201 destroys two of the three parameters. The *ATT* survives because it never needed that cell.
- This is common support failing.

Why it gets worse fast

- One binary covariate: 2 strata. Ten binary covariates: $2^{10} = 1,024$. One continuous covariate: infinitely many.
- Cells multiply exponentially; your sample does not. Empty cells become the norm, not the exception.
- Non-parametric methods **break down honestly**: they simply cannot compute the estimate.
- Regression **does not break down**: it extrapolates across empty cells using functional form, and reports a number with a standard error.

5. Exact Matching

Matching as explicit imputation

- Every causal inference is an imputation problem: the counterfactual is missing and must be filled in.
- Most methods impute **implicitly**, inside a formula. Matching does it **explicitly**, unit by unit — its main virtue.
- Start from the *ATT*. Exactly **one** term is unavailable:

$$ATT = \underbrace{E[Y^1 \mid D=1]}_{\text{observed: } Y=Y^1 \text{ when } D=1} - \underbrace{E[Y^0 \mid D=1]}_{\text{missing}}$$

- The first term is the treated sample mean. Matching builds a stand-in for the second.

Replacing the missing term

Condition on X , and the assumption does its work:

$$\begin{aligned} E[Y^0 \mid D=1, X=x] &= \underbrace{E[Y^0 \mid D=0, X=x]}_{\text{weak unconfoundedness}} \\ &= \underbrace{E[Y \mid D=0, X=x]}_{\text{switching equation}} \end{aligned}$$

$$E[Y^0 \mid D=1] = \underbrace{\sum_x E[Y \mid D=0, X=x] \Pr(X=x \mid D=1)}_{\text{common support: this must exist wherever treated units live}}$$

- The missing quantity is now written entirely in terms of control units.
- **Matching is simply a non-parametric estimate of $E[Y \mid D=0, X=X_i]$** — take the controls that look like i , and average them.

The estimator, and what to do about ties

Let $\mathcal{J}(i)$ be the set of control units matched to treated unit i , and $M_i = |\mathcal{J}(i)|$.

$$\widehat{ATT} = \frac{1}{N_T} \sum_{i: D_i=1} \left(Y_i - \underbrace{\frac{1}{M_i} \sum_{j \in \mathcal{J}(i)} Y_j}_{\widehat{Y}_i^0} \right)$$

- If two controls sit at the same distance from i , use both — breaking ties arbitrarily injects the analyst's coin flip. <!-- - M_i also varies when a caliper binds: a treated unit with no control inside the caliper contributes nothing, and N_T must then count only the units you actually matched.
- You can *point at the specific firms* doing the work for each observation. No regression output permits that. ->

The same idea for the ATE

For a control unit j , the missing half is Y^1 . Let $\mathcal{I}(j)$ be the treated units matched to it, $M_j = |\mathcal{I}(j)|$.

$$\widehat{ATE} = \frac{1}{N} \left[\underbrace{\sum_{i: D_i=1} \left(Y_i - \frac{1}{M_i} \sum_{j \in \mathcal{I}(i)} Y_j \right)}_{\text{treated: observed} - \text{imputed}} + \underbrace{\sum_{j: D_j=0} \left(\frac{1}{M_j} \sum_{i \in \mathcal{I}(j)} Y_i - Y_j \right)}_{\text{control: imputed} - \text{observed}} \right]$$

- Every unit contributes one *estimated* treatment effect; the *ATE* averages all N of them.
- Equivalently $\widehat{ATE} = \frac{N_T}{N} \widehat{ATT} + \frac{N_C}{N} \widehat{ATU}$ — the weights from §4, Subclassification, back again.

Exact matching and its limit

- With discrete X and full common support, exact matching and subclassification give the **same answer** — the same estimator, organized differently.
- The difference is bookkeeping: subclassification aggregates to stratum effects first; matching imputes unit by unit.
- Same fatal problem: with a continuous covariate, **exact matches occur with probability zero**.
- Two responses, and they define the next two sections:
 - ▶ Give up on exact equality — match to the *nearest* unit (§6), then correct that error (§8).
 - ▶ Give up on matching X itself — match on a *summary* of X (Lecture 3b, §1, Propensity Scores).

6. Nearest-Neighbour Matching

One confounder

- Drop the demand for exact equality. Match each treated unit to the control that minimizes $|X_i - X_j|$.
- The gap you accept is the **matching discrepancy**; §8 is about what it costs.
- Choices you must make and report:
 - ▶ **With or without replacement.** With replacement always takes the closest match, so it **lowers bias** and raises **variance**.
 - ▶ **How many neighbours, M .** More neighbours: less variance, more bias.
 - ▶ **Caliper:** refuse any match beyond a maximum distance. Better to drop a treated unit than match it to something unlike it.

Multiple dimensions and the distance metric

With several covariates, “closest” requires a definition — and the choice changes the answer.

- **Euclidean:** $\sqrt{\sum_k (X_{ik} - X_{jk})^2}$. Scale-dependent: a variable in won swamps one measured as a ratio.
- **Normalized Euclidean:** divide each covariate by its standard deviation. Fixes scale, ignores correlation.
- **Mahalanobis:** $\sqrt{(X_i - X_j)' \Sigma^{-1} (X_i - X_j)}$. Standardizes scale **and** accounts for correlation between covariates. The default choice.
- If size and leverage are highly correlated, Euclidean distance double-counts what is essentially one dimension.

7. Inference with Matching

Why the usual standard errors fail

- Matched observations are **dependent by construction**: the same control may serve several treated units, and matches were chosen *using the data*.
- Conventional standard errors treat the matched sample as randomly drawn. It was not, and they understate uncertainty.
- Abadie and Imbens (2008) show conventional standard errors are invalid for matching estimators.
- Use the **Abadie–Imbens variance estimator**, which prices in how many times each control was reused.
- Check what your software reports *by default*: several packages give the naive standard error unless told otherwise.

The Abadie–Imbens variance estimator

Let $\hat{\tau}_i = Y_i - \frac{1}{M} \sum_{m=1}^M Y_{j_m(i)}$ be unit i 's **own** effect, so $\widehat{ATT} = \frac{1}{N_T} \sum_{i:D_i=1} \hat{\tau}_i$ is their average. K_j counts how often control j is used.

$$\hat{\sigma}_{ATT}^2 = \underbrace{\frac{1}{N_T} \sum_{i:D_i=1} (\hat{\tau}_i - \widehat{ATT})^2}_{\text{(a) spread of the unit-level effects}} + \underbrace{\frac{1}{N_T} \sum_{j:D_j=0} \frac{K_j(K_j-1)}{M^2} \hat{\sigma}^2(X_j)}_{\text{(b) penalty for reusing controls}}$$

$$\text{SE}(\widehat{ATT}) = \sqrt{\hat{\sigma}_{ATT}^2 / N_T}$$

- Term (a) is the **sample variance of the** $\hat{\tau}_i$. Alone it is the without-replacement formula: with $K_j \in \{0, 1\}$, term (b) vanishes.
- $\hat{\sigma}^2(X_j) = \text{var}(Y \mid X_j, D=0)$, the outcome variance among controls.
- Here M is **fixed** — the clean $K_j(K_j-1)$ split needs it.

In Stata

Cunningham's training data: 10 trainees, 20 non-trainees, matched on age and gpa.

```
use https://github.com/scunning1975/mixtape/raw/master/training\_inexact.dta, clear

* Minimized euclidean distance with usual variance and robust
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(eucl) vce(iid)
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(eucl) generate(match1)

* Minimized Mahalanobis distance with usual variance and robust
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha) vce(iid)
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha) generate(match2)
```

- (earnings age gpa) is outcome-then-covariates; (treat) is the treatment. atet asks for the *ATT*, nn(1) for one neighbour.
- vce(iid) is the constant-variance standard error. **Drop it and you get Abadie–Imbens**, Stata's default — so report the second command of each pair.
- generate(match1) writes each unit's matched partner back into the data. Always look at it.

Two metrics, two answers

	$\widehat{ATT}$	vce(iid)	Abadie–Imbens	distinct controls used
Euclidean	1607.5	38.7	38.7	10 of 20
Mahalanobis	1315.0	325.5	590.1	3 of 20
<i>Truth</i>	1607.5			

- age has standard deviation 9.1, gpa only 0.57, so raw Euclidean distance **matches on age and ignores gpa** — the scale problem from §6. Every trainee gets a *different* control: $K_j \leq 1$, term (b) is zero, and both standard errors coincide.
- Mahalanobis makes gpa count — and gpa is noise. **Three controls serve all ten trainees**; one is used five times.
- Term (a) alone still reports 199.8. The reuse correction triples it.

8. Bias Correction

The matching discrepancy is bias, not noise

Matching was only *approximate*: $X_{j(i)} \approx X_i$, never $=$. What does the leftover gap cost?

- Abadie–Imbens (2006): More data does not help much.
- Usually you cannot get more data anyway. Abadie–Imbens (2011) is the second-best answer: **model the bias and subtract it.**

An outcome model for the missing Y^0

$$\begin{aligned}\mu^0(x) &= E[Y \mid X=x, D=0] = \underbrace{E[Y^0 \mid X=x]}_{\text{unconfoundedness}} \\ \mu^1(x) &= E[Y \mid X=x, D=1] = \underbrace{E[Y^1 \mid X=x]}_{\text{unconfoundedness}}\end{aligned}$$

$$Y_i = \mu^{D_i}(X_i) + \varepsilon_i, \quad E[\varepsilon_i \mid X_i, D_i] = 0$$

- The second equality on each line **is** unconfoundedness: it lets a regression fitted on controls speak about treated units.
- So far, all definition. The **assumption** arrives only when we choose a functional form for $\hat{\mu}^0$.

Where the bias hides

Substitute $Y_i = \mu^{D_i}(X_i) + \varepsilon_i$ into $\widehat{ATT} = \frac{1}{N_T} \sum_{D_i=1} (Y_i - Y_{j(i)})$, subtract the true ATT , then add and subtract $\frac{1}{N_T} \sum_{D_i=1} \mu^0(X_i)$:

$$\begin{aligned}\widehat{ATT} - ATT &= \underbrace{\frac{1}{N_T} \sum_{D_i=1} (\mu^1(X_i) - \mu^0(X_i) - ATT)}_{(1) \text{ sampling variation in } \textit{which} \text{ units got treated}} \\ &+ \underbrace{\frac{1}{N_T} \sum_{D_i=1} (\varepsilon_i - \varepsilon_{j(i)})}_{(2) \text{ mean-zero noise}} \\ &+ \underbrace{\frac{1}{N_T} \sum_{D_i=1} (\mu^0(X_i) - \mu^0(X_{j(i)}))}_{(3) \text{ matching bias}}\end{aligned}$$

- Scaled by $\sqrt{N_T}$, rows (1) and (2) converge to mean-zero normals: noise, and they average away.
- Row (3) does not. Its summands are **signed**: if the nearest control is systematically older or smaller, every term carries the same sign.

Two things make the bias big

$$\text{Matching bias} = E[\mu^0(X_i) - \mu^0(X_{j(i)}) \mid D = 1]$$

- Read it as a product of two quantities, and it is zero if **either** is:
 - ▶ **How bad the matches are** — the size of $X_i - X_{j(i)}$. Perfect matching kills it.
 - ▶ **How steep μ^0 is** — the heterogeneity of Y^0 across strata of X . A flat μ^0 kills it too.
- Match a 13-year-old to a 15-year-old and nothing is lost *unless* Y^0 differs between 13- and 15-year-olds.
- For the *ATT* it is Y^0 that is missing and imputed, so only μ^0 has to be modelled.

The Abadie–Imbens procedure, in five steps

1. **Match.** Nearest neighbour on your metric; form $\hat{\delta}_i = Y_i - Y_{j(i)}$ for each treated unit.
2. **Regress.** OLS on the **matched control units only**: $Y_j = \alpha + \beta X_j + e_j$.
3. **Predict.** $\hat{\mu}^0(X_i) = \hat{\alpha} + \hat{\beta}X_i$ and $\hat{\mu}^0(X_{j(i)}) = \hat{\alpha} + \hat{\beta}X_{j(i)}$.
4. **Subtract** the estimated bias:

$$\widehat{ATT}^{BC} = \frac{1}{N_T} \sum_{D_i=1} \left[(Y_i - Y_{j(i)}) - (\hat{\mu}^0(X_i) - \hat{\mu}^0(X_{j(i)})) \right]$$

5. **Report an Abadie–Imbens standard error** (§7, Inference with Matching). The correction changes the estimate, not the reuse problem.
- Step 2 never touches a treated outcome: $\hat{\mu}^0$ is fitted where Y^0 is *observed*, then transplanted. Unconfoundedness permits the transplant.

Only the gap has to be right

- The estimator uses only the **difference** $\hat{\mu}^0(X_i) - \hat{\mu}^0(X_{j(i)})$. With one covariate and a linear fit that is simply $\hat{\beta}(X_i - X_{j(i)})$.
- So $\hat{\mu}^0$ must be accurate *over the gap*, not globally. Get the level wrong everywhere and the correction is unaffected: $\hat{\alpha}$ cancels.
- **This is why bias correction is more credible than regression alone.** Regression extrapolates $\hat{\mu}^0$ across the whole covariate space; bias correction interpolates across a gap matching already made small.
- Division of labour: **matching does the heavy lifting on comparability; regression cleans up the remainder.**
- The worse your matches, the more work the regression does — and the more the method degenerates into regression.

Ten workers

Unit	Worker	Y^1	$Y^0 / \widehat{Y}^0$	$\hat{\delta}_i$	Age	D
1	Andy	\$200	\$195	\$5	23	1
2	Betty	\$250	\$225	\$25	27	1
3	Chad	\$150	\$140	\$10	29	1
4	Doug	\$300	\$195	\$105	22	1
5	Edith	—	\$225		27	0
6	Fred	—	\$500		32	0
7	Gina	—	\$200		26	0
8	Hank	—	\$190		26	0
9	Inez	—	\$180		28	0
10	Janet	—	\$140		29	0

- Andy (23) and Doug (22) both match Gina and Hank, the 26-year-olds:
 $\widehat{Y}^0 = \frac{200+190}{2} = 195$. Betty and Chad match exactly.
- $\widehat{ATT} = (5 + 25 + 10 + 105)/4 = \36.25 . Fred and Inez are never used.
- Both inexact matches point the same way: the match is *older* than the trainee.

Steps 2–4, worked through

Step 2, over the four *matched* controls — Edith (27, 225), Gina (26, 200), Hank (26, 190), Janet (29, 140): $\hat{\mu}^0(\text{age}) = 683.75 - 18.33 \text{ age}$.

Worker	Age	Match age	$\hat{\delta}_i$	$\hat{\beta}(X_i - X_{j(i)})$	$\hat{\delta}_i^{BC}$
Andy	23	26	\$5	\$55.0	-\$50.0
Betty	27	27	\$25	\$0	\$25.0
Chad	29	29	\$10	\$0	\$10.0
Doug	22	26	\$105	\$73.3	\$31.7

$$\widehat{ATT} = \$36.25 \longrightarrow \widehat{ATT}^{BC} = \$4.17.$$

- Exact matches get **zero** correction. Only the two bad matches move.
- $\hat{\beta} < 0$: among *these four* controls earnings fall with age. Refit on all six and Fred (32, \$500) flips it to $\hat{\beta} = +43.7$, giving $\widehat{ATT}^{BC} = \$112.6$.
- Four points and one outlier. The “small” regression is doing enormous work — the method’s exposure.

What the correction does and does not do

- It converts a *matching* problem into a *much easier regression* problem. It does not eliminate the regression.
- It is honest about its assumption: you must name a functional form for $\hat{\mu}^0$, and the ten-worker example shows how much that moves the answer.
- It does **not** repair the matches.
- Report the raw and bias-corrected estimates side by side. A large gap is a **diagnostic**: your matches were poor and the outcome model is load-bearing.

Wrapping Up

Summary [Part 1]: the assumption

- **Assignment mechanisms resolve bias, not models.** The SDO decomposition is an identity; only a claim about how treatment was assigned deletes its bias terms.
- **Covariates come in four kinds:** confounders (include), colliders (never), precision covariates (optional), mediators (never). Pre-treatment timing is your best protection.
- **Unconfoundedness**, $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$: randomization *within* strata of X . Untestable; defended institutionally, and Imbens's third defence is strongest in finance.
- **Common support** is a *separate* assumption, and unlike unconfoundedness it is **checkable**. It is asymmetric: the *ATT* needs only $e(X) < 1$, the *ATE* both ends.
- **Subclassification** makes the estimand a choice about weights. On the Titanic: $SDO = 35.4$ pp, but $ATE = 19.6$, $ATT = 23.8$, $ATU = 18.9$.
- The **curse of dimensionality** is structural, not a small-sample nuisance. One missing passenger killed two of three parameters.

Summary [Part 2]: matching, and what it costs

- **Matching** is explicit imputation: replace the missing Y_i^0 with a comparable control's observed outcome. Exact matching is the honest version, and it fails on continuous X .
- **Nearest-neighbour matching** buys feasibility with a **matching discrepancy** that is *signed*, not noise: treated units with high X are matched downward. It is bias.
- **The bias shrinks slower than the standard error:** $O_p(N^{-1/k})$ against $O_p(N^{-1/2})$. At $k = 2$ they are the same order; at $k \geq 3$ the bias dominates and the t -statistic is invalid.
- **Two separate repairs.** The *variance* estimator fixes inference: matching with replacement makes comparisons dependent, and $K_j(K_j - 1)$ prices that in. *Bias correction* fixes the estimate, using a regression only to price the gap $\hat{\mu}(X_i) - \hat{\mu}(X_{j(i)})$ — far weaker than modelling the level of Y .
- **Scale matters.** Euclidean and Mahalanobis are different estimators, not different implementations: on the training data they return 1607.5 and 1315.0.
- Next: the same assumption, executed by collapsing X to **one number**.

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